

PROPOSITION 1 (CAPACITY TRAP). Fix $\alpha > 0$, $\beta \geq 1$, $t_A < t_B$, and $0 < c_A < c_B$. There exists $p \in (0, 1)$ and $D_0 < \infty$ such that for every $D > D_0$,

$$C_{\text{split}}(D, p) < C_{\text{mono}}(D).$$

PROOF. The order- $D^{\beta+1}$ congestion coefficient of the split cost is

$$g(p) = t_A p^{\beta+1}/c_A^\beta + t_B(1-p)^{\beta+1}/c_B^\beta.$$

The corresponding coefficient of the monoculture cost is $g(1) = t_A/c_A^\beta$. The derivative from the left at $p = 1$ is

$$g'(1) = (\beta + 1)t_A/c_A^\beta > 0,$$

so decreasing p slightly below 1 reduces the leading congestion coefficient. Hence there is $p \in (0, 1)$ with $g(p) < g(1)$. For this p , the difference $C_{\text{mono}}(D) - C_{\text{split}}(D, p)$ has positive leading term

$$\alpha(g(1) - g(p))D^{\beta+1},$$

while the free-flow penalty from moving $(1-p)D$ units to the slower link is order D . Since $\beta \geq 1$, the congestion gain eventually dominates the linear free-flow penalty. Therefore some finite D_0 satisfies the claimed inequality for all $D > D_0$. $\square$

The proposition is intentionally constructive. It does not say that monoculture is always harmful; it says that a low-free-time, low-capacity shortcut can become a trap when advice is correlated. Writing $a = t_A/c_A^\beta$ and $b = t_B/c_B^\beta$, the high-demand coefficient is $g(p) = ap^{\beta+1} + b(1-p)^{\beta+1}$. Its first-order condition is

$$ap^\beta = b(1-p)^\beta,$$

so the coefficient is minimized at

$$p^\star = \frac{(t_B/c_B^\beta)^{1/\beta}}{(t_A/c_A^\beta)^{1/\beta} + (t_B/c_B^\beta)^{1/\beta}},$$

which assigns less mass to the low-capacity shortcut when c_A is small. The exact finite-demand optimum also includes the free-flow terms, but the leading coefficient explains why diversified flow can dominate at high load.

PROPOSITION 2 (CAPACITY-PROPORTIONAL SUFFICIENT CONDITION). Let $p_c = c_A/(c_A + c_B)$. If

$$\frac{t_A}{c_A^\beta} > \frac{t_A c_A + t_B c_B}{(c_A + c_B)^{\beta+1}},$$

then there exists $D_0 < \infty$ such that $C_{\text{split}}(D, p_c) < C_{\text{mono}}(D)$ for all $D > D_0$.

PROOF. For the capacity-proportional split, the order- $D^{\beta+1}$ congestion coefficient is

$$g(p_c) = \frac{t_A c_A + t_B c_B}{(c_A + c_B)^{\beta+1}}.$$

The displayed condition is exactly $g(p_c) < g(1)$. The same leading-term comparison used in the capacity-trap proposition gives the result. $\square$

The capacity-proportional condition is only sufficient. The optimal split for a finite instance may be neither capacity-proportional nor exactly $p^\star$, and a network with several overlapping paths can have more complicated bottleneck structure. Its value is interpretive: it makes explicit that a small free-flow advantage on a low-capacity edge can be dominated by convex congestion.

Finite-demand threshold. The asymptotic argument also yields a directly checkable sufficient condition. For any fixed split with $g(p) < g(1)$,

$$C_{\text{mono}}(D) - C_{\text{split}}(D, p) = D(1-p)(t_A - t_B) + \alpha D^{\beta+1}(g(1) - g(p)).$$

Thus a split is cheaper whenever

$$D^\beta > \frac{(1-p)(t_B - t_A)}{\alpha(g(1) - g(p))}.$$

The threshold is sufficient rather than tight, but makes the high-load mechanism auditable at finite demand.

Cost decomposition. Equivalently, the cost difference separates the free-flow penalty from the congestion gain:

$$\begin{aligned} C_{\text{mono}}(D) - C_{\text{split}}(D, p) &= D(1-p)(t_A - t_B) \\ &\quad + \alpha D^{\beta+1} \left[\frac{t_A(1-p^{\beta+1})}{c_A^\beta} - \frac{t_B(1-p)^{\beta+1}}{c_B^\beta} \right]. \end{aligned}$$

The first term is non-positive because assigning flow to B sacrifices free-flow time. The second is positive exactly when the split reduces the leading congestion coefficient, clarifying why sufficiently high demand reverses the free-flow ranking.

Partial compliance. Let $q \in [0, 1]$ be the fraction of demand that follows the shared shortcut recommendation. The remaining users follow a diversified fallback that sends fraction p_f of fallback flow to A . Then

$$x_A = D(q + (1-q)p_f), \quad x_B = D(1-q)(1-p_f),$$

and

$$C_{\text{partial}}(D, q, p_f) = x_A t_A \left(1 + \alpha(x_A/c_A)^\beta\right) + x_B t_B \left(1 + \alpha(x_B/c_B)^\beta\right).$$

Writing $s(q) = p_f + q(1-p_f)$, the leading congestion coefficient is

$$h(q) = t_A s(q)^{\beta+1}/c_A^\beta + t_B(1-s(q))^{\beta+1}/c_B^\beta.$$

Its derivative is

$$h'(q) = (1-p_f)(\beta+1) \left(t_A s(q)^\beta/c_A^\beta - t_B(1-s(q))^\beta/c_B^\beta \right).$$

For $p_f < 1$, the derivative has the sign of the parenthesized difference; when $p_f = 1$, it is zero because compliance no longer changes the shortcut share. Thus, in a high-load capacity-trap regime, increasing q raises leading congestion once the shortcut already carries enough mass. This monotonicity is not global: at low demand, or when the shortcut is not capacity constrained, higher compliance can have little effect or can reduce free-flow travel time. The experiments therefore report partial-compliance trends as empirical evidence, not as a universal proposition.